Practical 3c: BankAccount Class with Property Validation

AIM : To create a BankAccount class with AccountNumber and Balance properties, where the Balance property includes validation to prevent negative balances.

PROGRAM CODE

```
using System;

namespace BankAccountApp
{
   class BankAccount
   {
      private string accountNumber;
      private double balance;

      public string AccountNumber
      {
         get { return accountNumber; }
         set
         {
            if (string.IsNullOrEmpty(value))
               throw new ArgumentException("Account number cannot be empty!");
            accountNumber = value;
         }
      }

      public double Balance
      {
         get { return balance; }
         set
         {
            if (value < 0)
               throw new ArgumentException("Balance cannot be negative!");
            balance = value;
         }
      }

      public BankAccount(string accNo, double initialBalance)
      {
```

```
        AccountNumber = accNo;
        Balance = initialBalance;
    }

    public void Deposit(double amount)
    {
        if (amount <= 0) throw new ArgumentException("Deposit must be positive.");
        Balance += amount;
        Console.WriteLine($"Deposited Rs.{amount}. New Balance: Rs.{Balance}");
    }

    public void Withdraw(double amount)
    {
        if (amount > Balance) throw new InvalidOperationException("Insufficient funds!");

        Balance -= amount;
        Console.WriteLine($"Withdrawn Rs.{amount}. New Balance: Rs.{Balance}");
    }

    public void DisplayInfo()
    {
        Console.WriteLine($"Account: {AccountNumber} | Balance: Rs.{Balance}");
    }
}

class Program
{
    static void Main(string[] args)
    {
        Console.WriteLine("=== Bank Account Demo ===");
        BankAccount acc = new BankAccount("ACC1001", 5000);
        acc.DisplayInfo();
        acc.Deposit(2000);
        acc.Withdraw(1500);
        acc.DisplayInfo();

        Console.WriteLine("\nTrying negative balance:");
        try { acc.Balance = -100; }
        catch (Exception ex) { Console.WriteLine($"Error: {ex.Message}"); }
```

```
        Console.WriteLine("\nTrying overdraft:");
        try { acc.Withdraw(50000); }
        catch (Exception ex) { Console.WriteLine($"Error: {ex.Message}"); }

        Console.ReadKey();
    }
  }
}
```

OUTPUT

```
=== Bank Account Demo ===
Account: ACC1001 | Balance: Rs.5000
Deposited Rs.2000. New Balance: Rs.7000
Withdrawn Rs.1500. New Balance: Rs.5500
Account: ACC1001 | Balance: Rs.5500

Trying negative balance:
Error: Balance cannot be negative!

Trying overdraft:
Error: Insufficient funds!
```